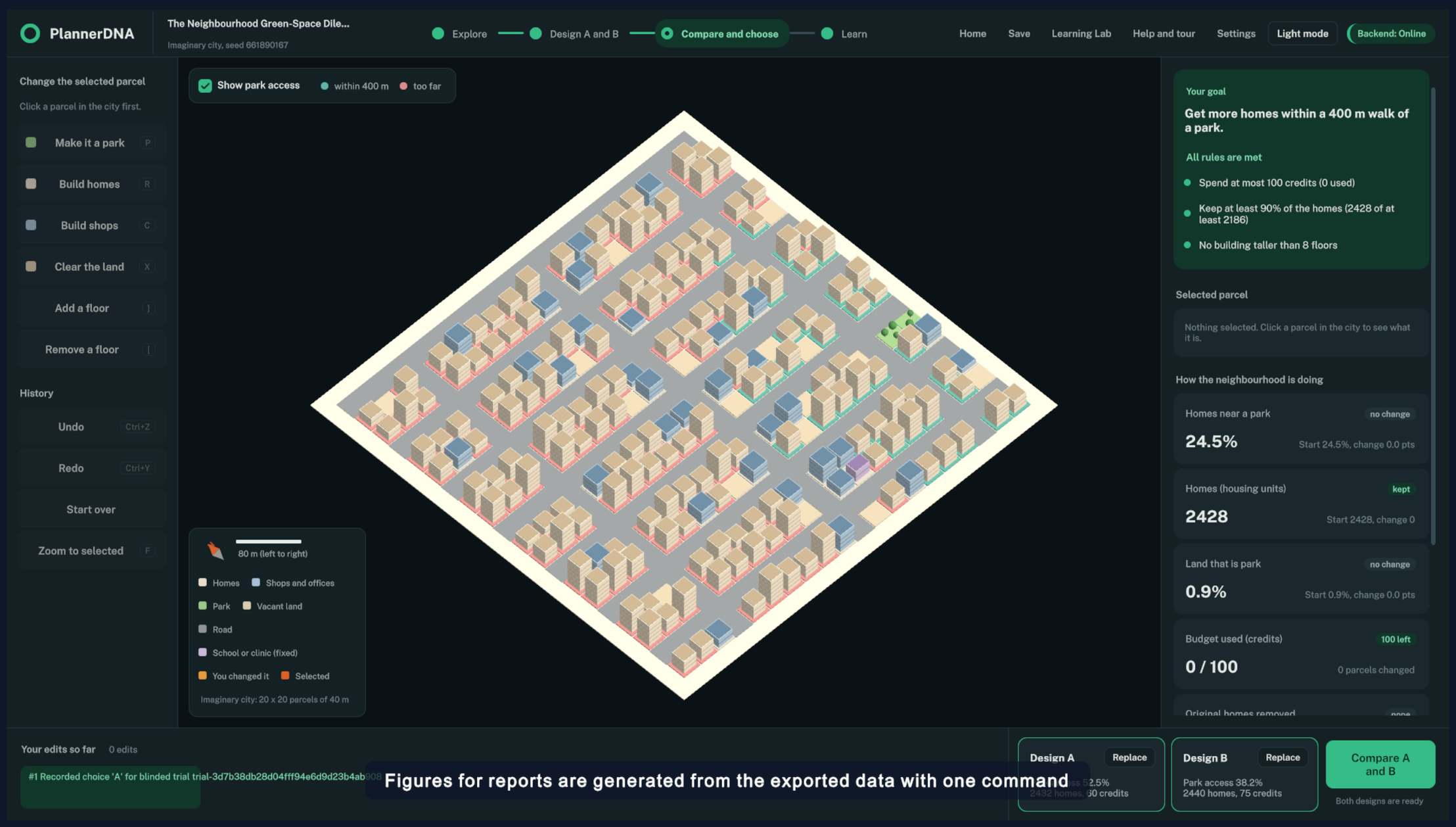
PlannerDNA

A 3D urban-design sandbox that learns from a student's explicit A/B planning choices, and then tests that model fairly.



Unity 6 | FastAPI + SQLite | scikit-learn | runs entirely on one computer

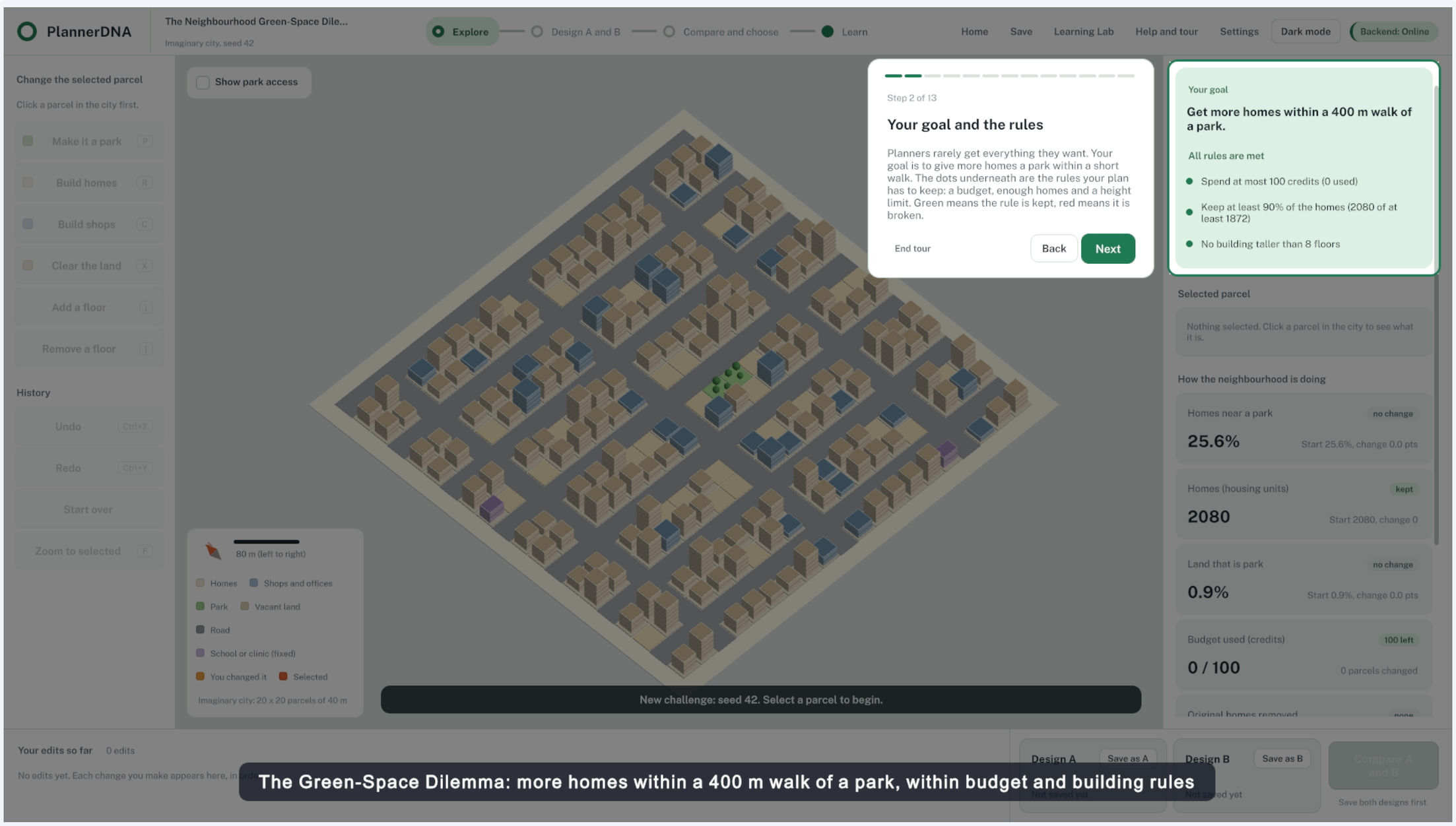
Synthetic city and synthetic metrics. A learning and research tool, not planning advice.

THE IDEA

Make the trade-offs of planning visible, then ask what a person actually prefers

The Green-Space Dilemma: give more homes a park within a 400 m walk, while keeping at least 90% of the housing, spending at most 100 credits and building no higher than 8 floors.

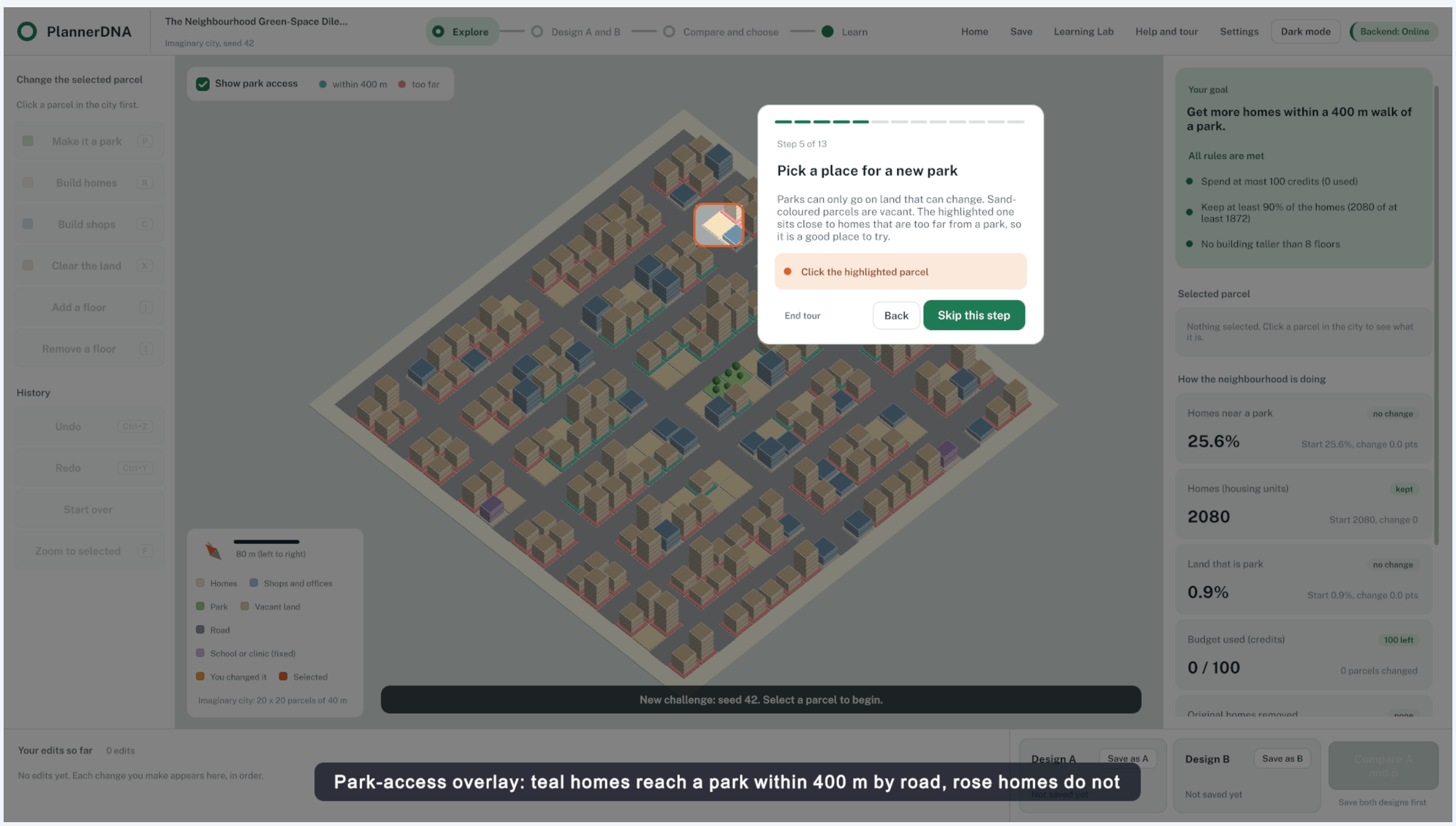
- Explore a generated 20 x 20 neighbourhood.
- Design A and B from the same starting point.
- Compare and choose: every explicit choice is a data point.
- Learn: a small, explainable model summarises the choices, only when asked.



THE SANDBOX

Transparent rules, recomputed on every edit

- Park access: share of homes with a park within 400 m along the road network (breadth-first search).
- Housing capacity: floors x units per floor; park share of non-road land.
- Cost from the difference to the baseline, so restoring a parcel always costs 0.
- Hard rules shown as green or red dots. Exact undo and redo; edits are logged, never treated as preferences.

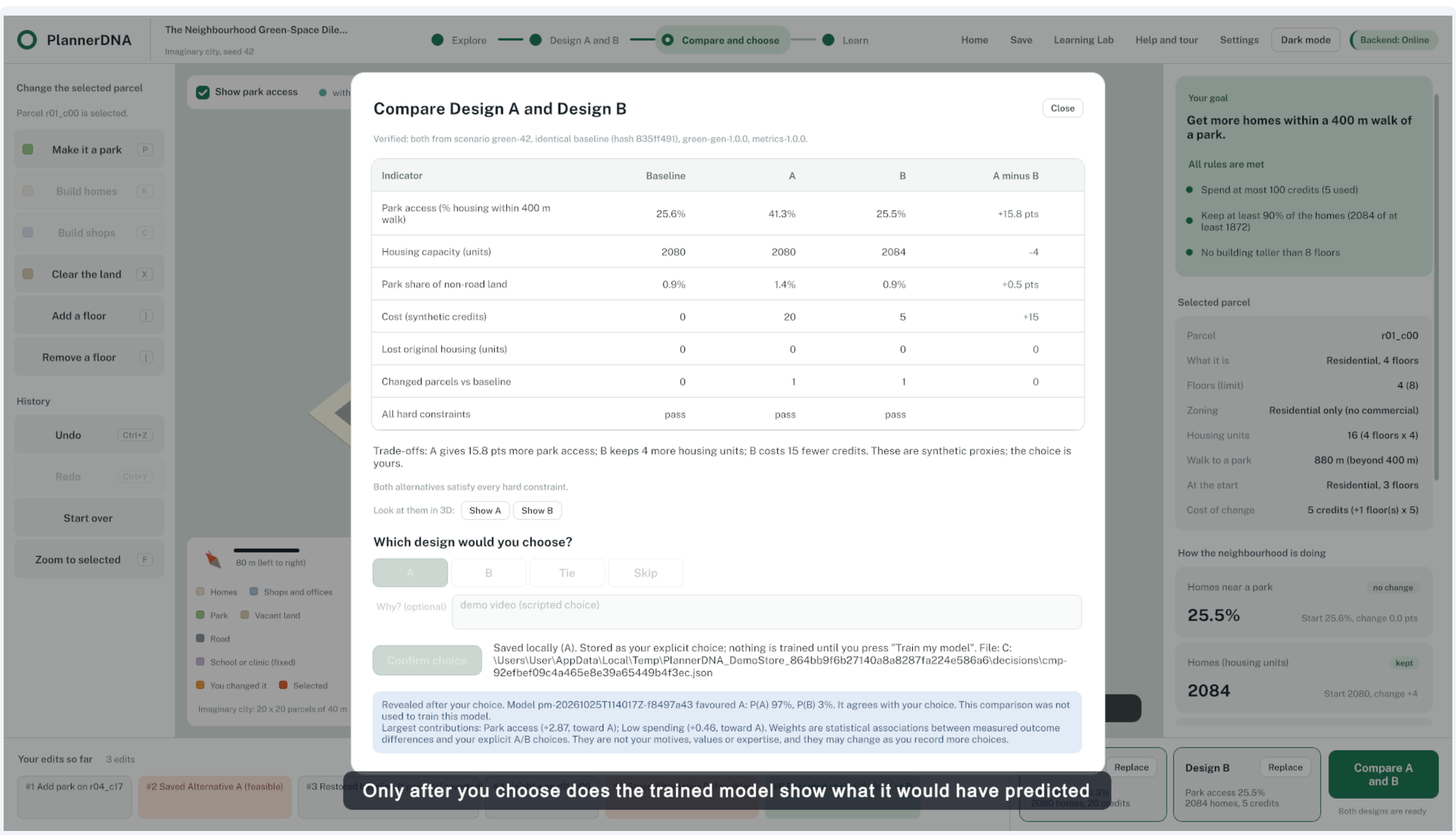


A 13-step guided tour points at each part of the screen and waits for the student to do the real action. Written for first-year urban-planning undergraduates.

COMPARE AND CHOOSE

The prediction is shown only after the choice

Both designs are checked against the same baseline. The student picks A, B, tie or skip and may type a reason. Only an explicit A or B choice between two feasible designs counts as a training example.



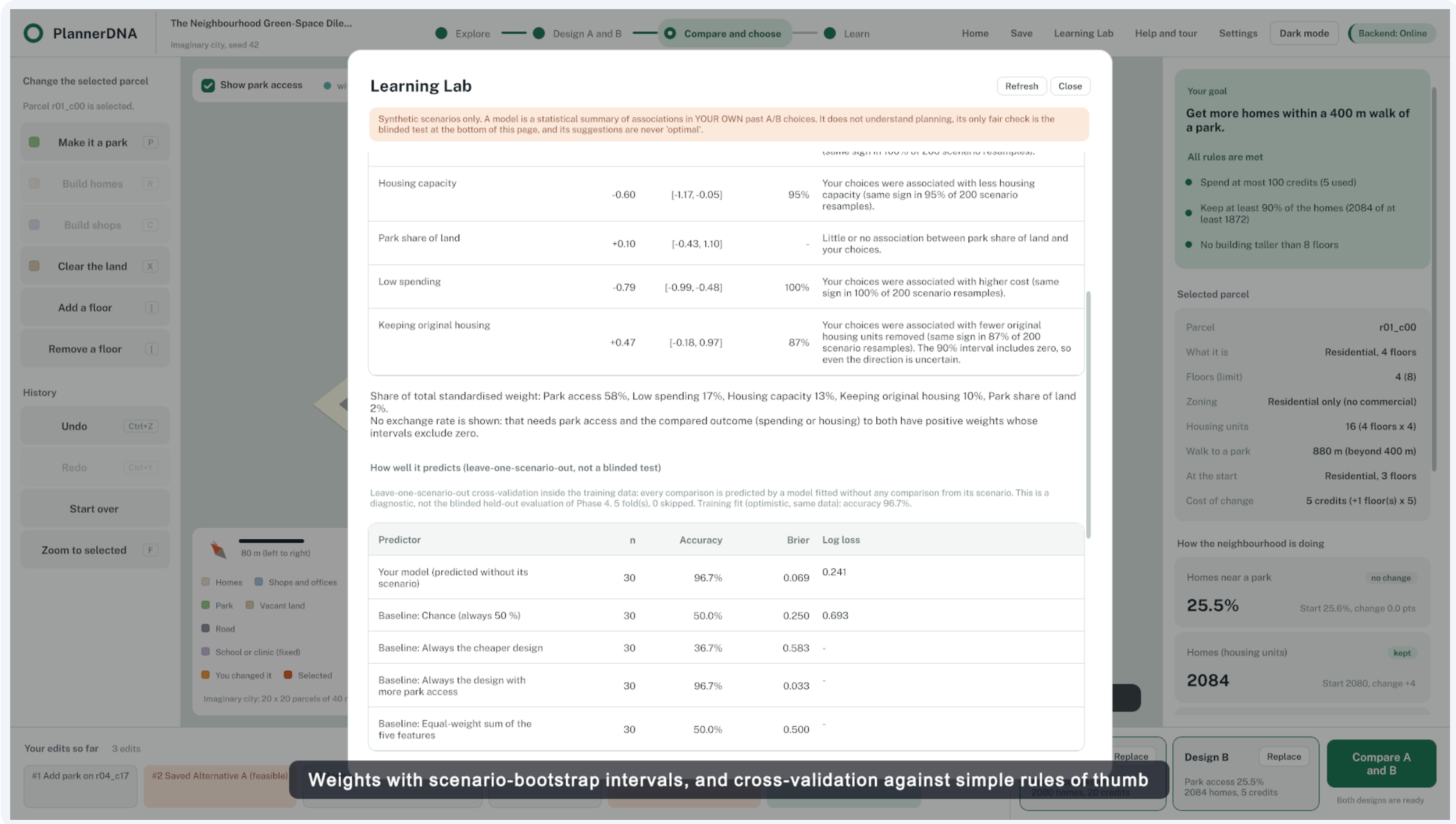
Showing the model's guess first would anchor the choice. Here it is revealed afterwards, with the outcomes that drove it and whether the pair was in the training data.

THE MODEL

Pairwise logistic regression: small enough to explain line by line

P(A chosen) = sigmoid(w · [f(A) − f(B)]), with five outcome differences f: park access, housing capacity, park share of land, low spending, keeping original housing.

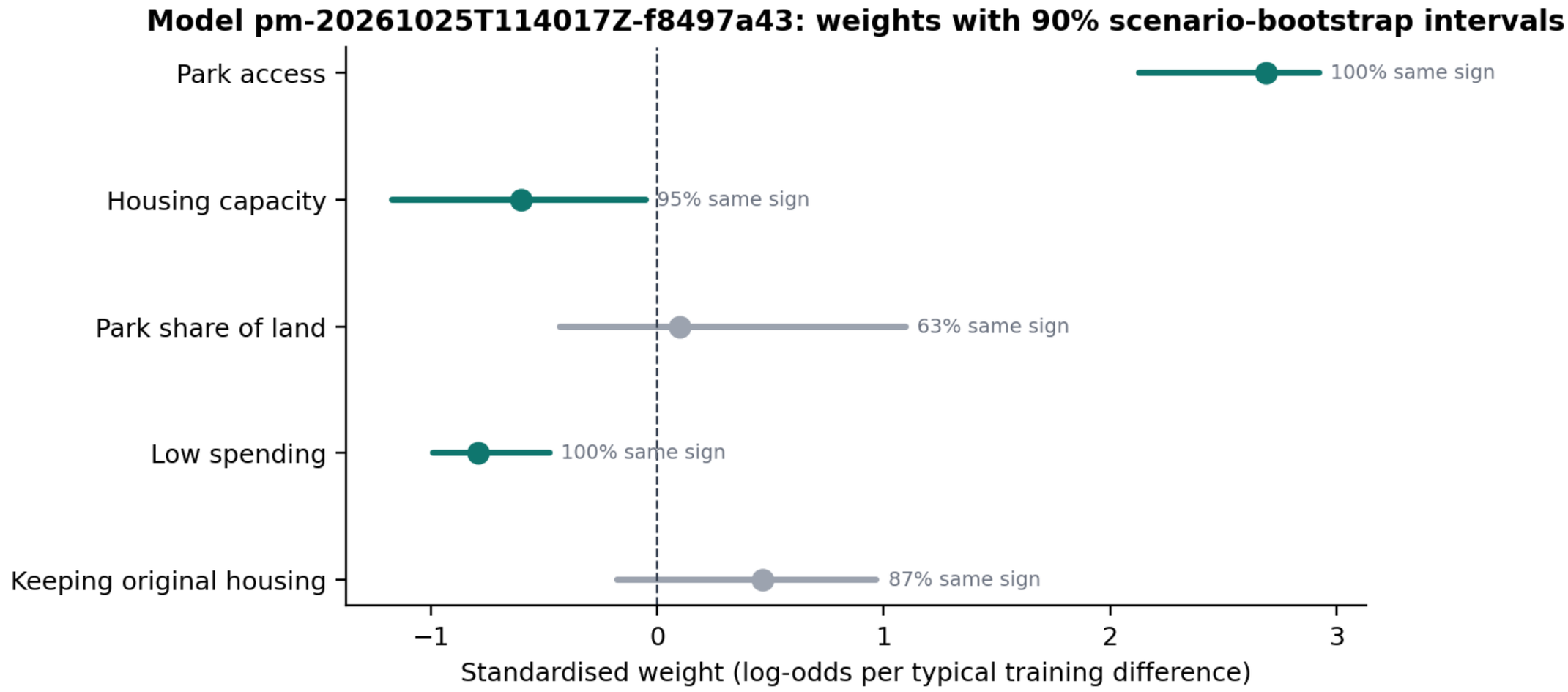
- No intercept, and every choice is also added with A and B swapped: screen position cannot be learned.
- L2-regularised (C = 1), standardised features, fixed seed, trained only on an explicit button press.
- 200 scenario-bootstrap refits give 90% intervals; leave-one-city-out cross-validation is compared with rules of thumb.
- Stored as checksummed JSON (never pickle); identical data always give the identical model.



Weights with scenario-bootstrap intervals, and cross-validation against simple rules of thumb

WHAT IT PICKED UP

Weights are associations with choices, not motives



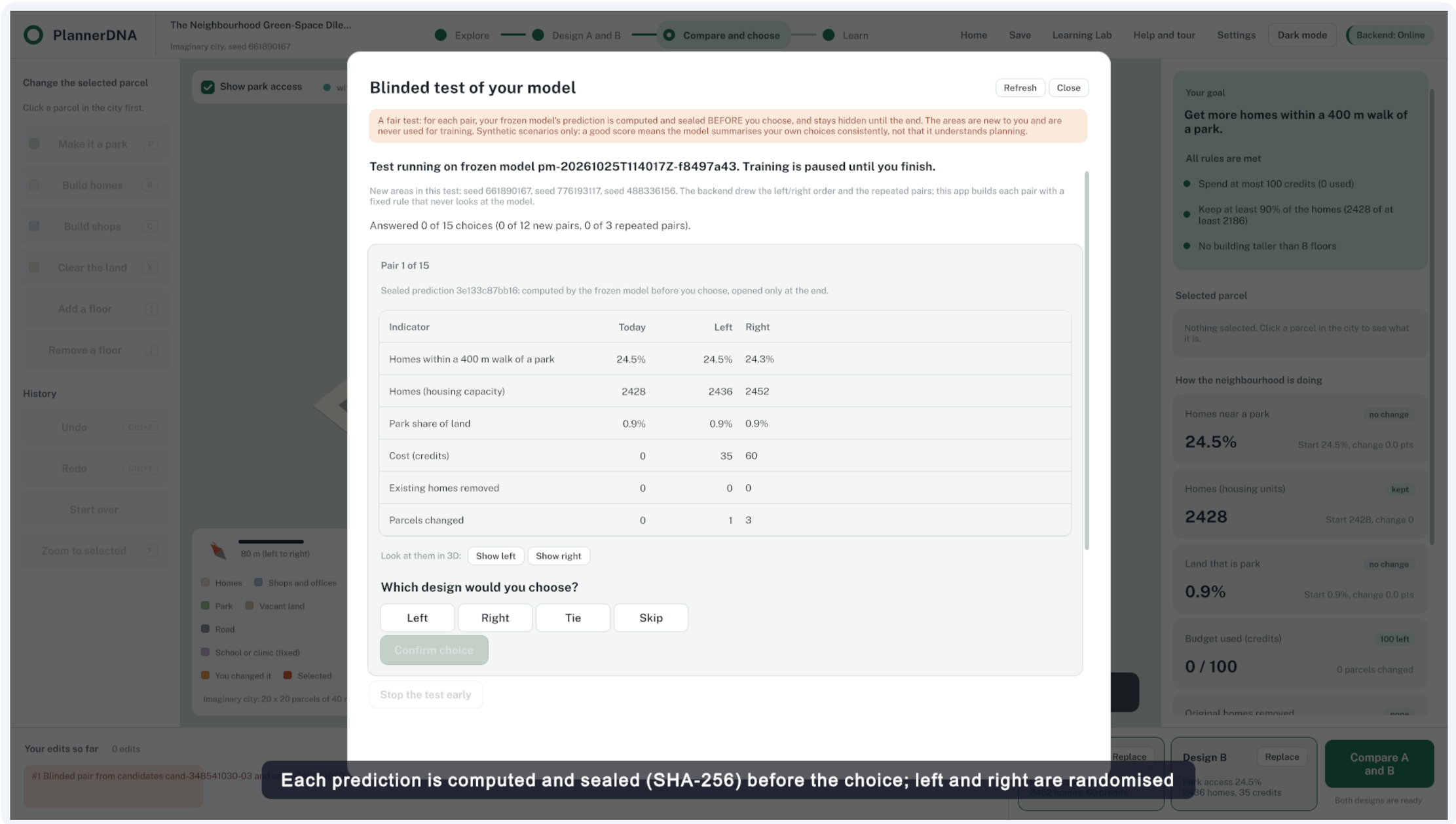
Trained on 30 comparisons across 5 synthetic cities. Park access dominates (standardised weight 2.69, 90% interval 2.13 to 2.93). Leave-one-city-out accuracy 96.7%, the same as the rule "always more park access" (96.7%).

Data note: 10 of these comparisons were made by me; 20 were generated by a fixed rule ("more park access, then lower cost") to complete the demo, which is why the figures say so.

THE FAIR TEST

A pre-registered, blinded evaluation

- Freeze the model by version and file checksum; training is refused until the test ends.
- New areas only: cities never trained on or practised, checked on the server.
- Seal before the choice: the server stores the prediction and shows only its SHA-256 fingerprint.
- Randomise left and right; repeat some pairs later with the sides swapped.
- Reveal at the end: every seal is checked; the report is stored with its hash and recomputed on read.



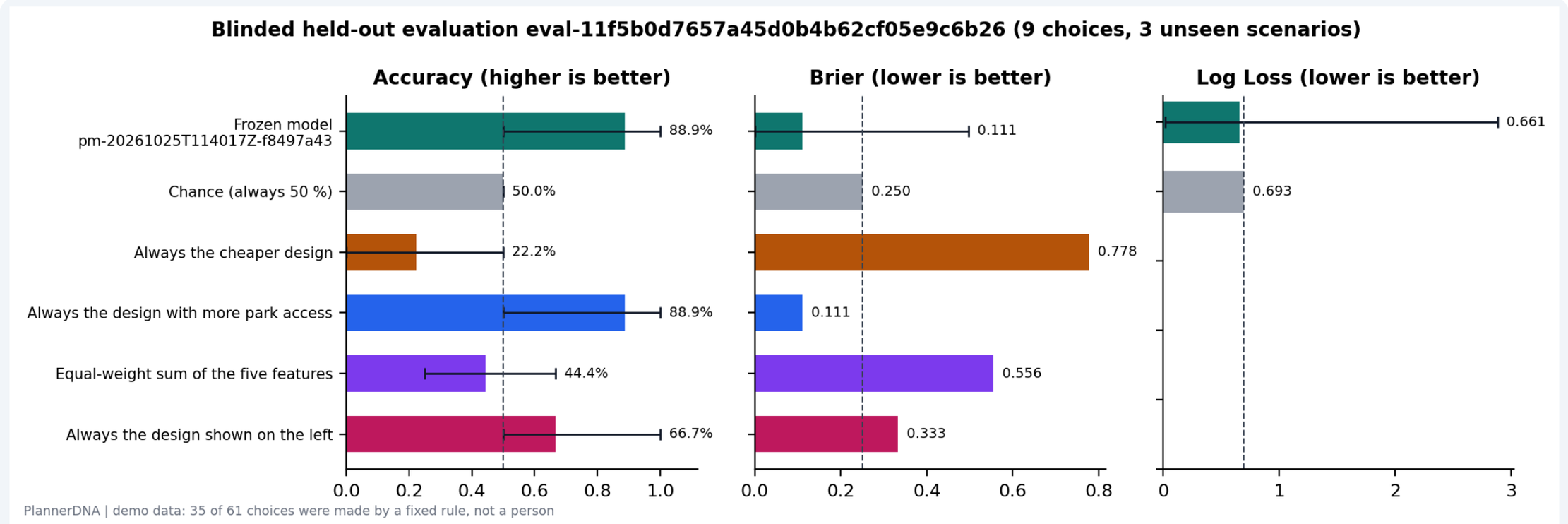
RESULT OF MY BLINDED TEST

It matched my choices, but so did a one-line rule

8 / 9
held-out choices predicted by the frozen model

88.9%
rule: always the design with more park access

50.0%
chance

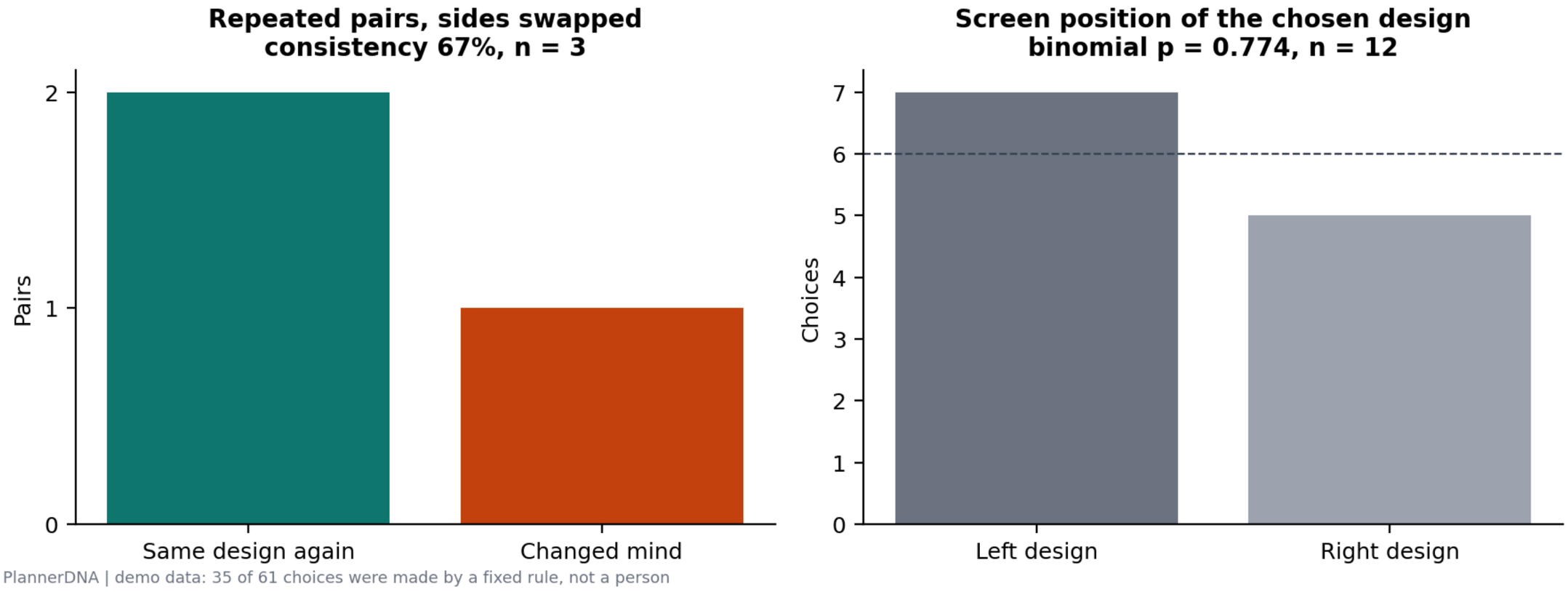


15 answers in 3 new cities: 9 scored A/B choices (3 ties and 3 repeats are not scored). Model accuracy 88.9%, 95% interval 50.0% to 100.0% (bootstrap over whole cities); Brier 0.111, log loss 0.661.

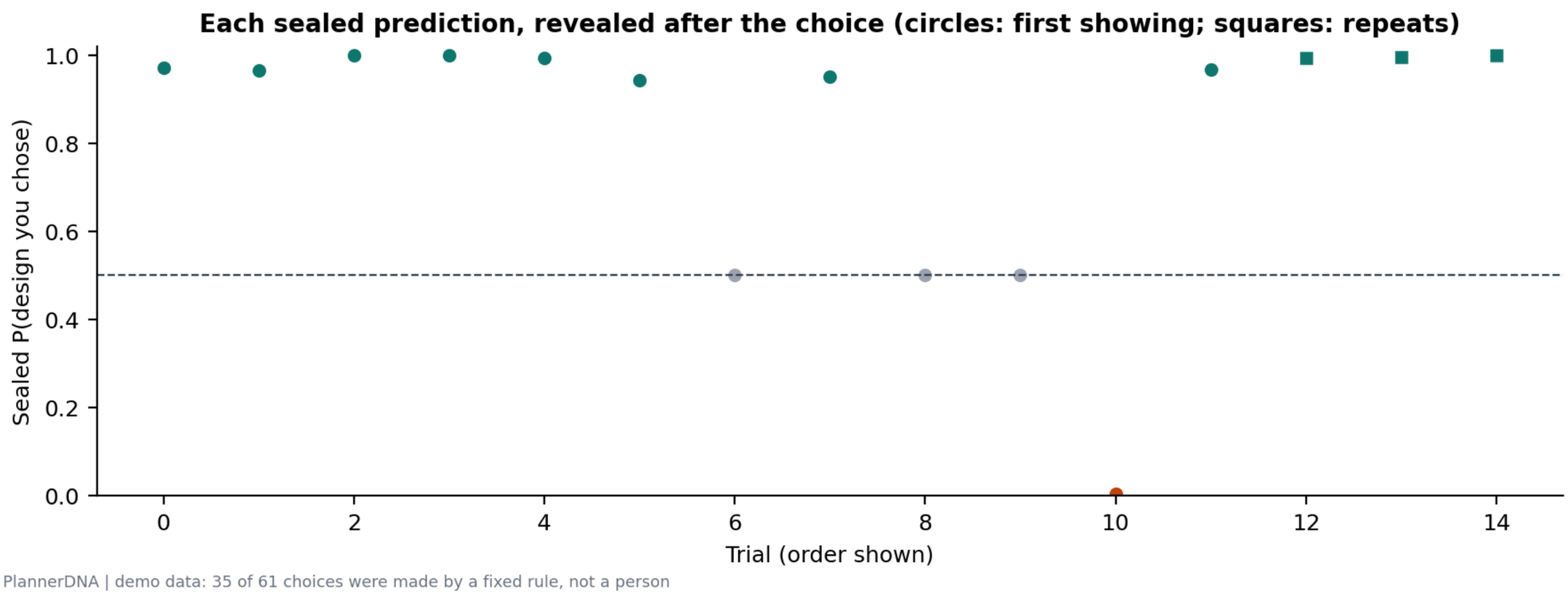
Honest reading: with this little data, the model has not shown that it captures more than "prefer more park access".

HOW CONSISTENT WAS I?

Inconsistent choices cap any model's accuracy



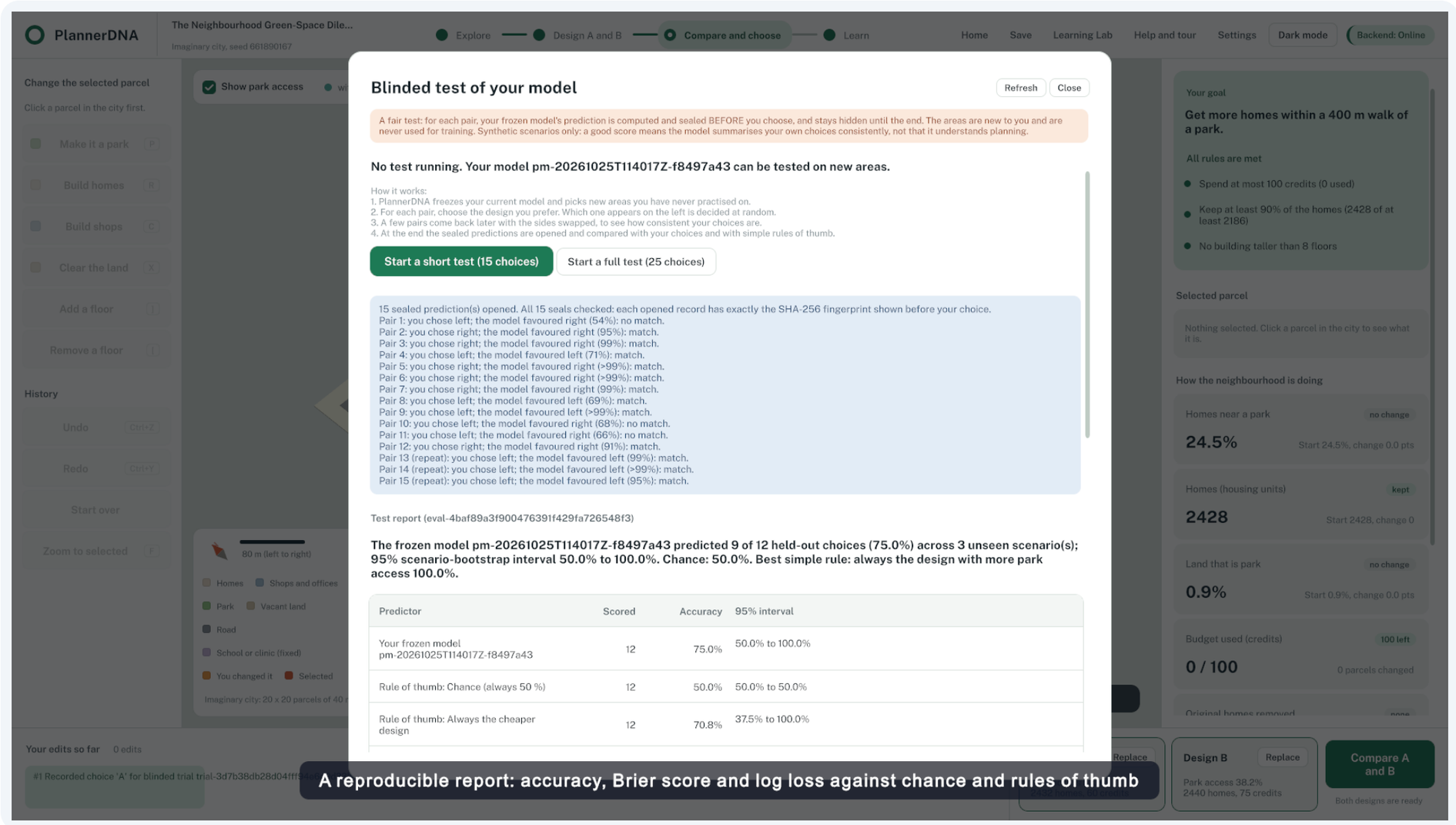
Repeated pairs (sides swapped): same choice in 2 of 3. Left-hand design chosen in 7 of 12 (binomial p = 0.77: no evidence of a side bias).



ENGINEERING

Offline-first, reproducible, tested

- Unity 6 (URP, UI Toolkit, Input System): immutable city state, deterministic SplitMix64 generation, snapshot undo/redo, light and dark themes.
- Local backend: FastAPI + SQLite with migrations, foreign keys and transactions; binds to 127.0.0.1 only.
- Every record is written locally first and synced in order through an outbox with idempotent retries.
- The server recomputes every metric and hash from the snapshot and refuses any mismatch.
- Figures are generated from exported data with the numbers in a CSV beside each one; the model is refitted from the export to check it reproduces exactly.
- Automated tests: 88 backend, 130 Unity EditMode and 24 Unity PlayMode tests (incl. a full blinded test).



A reproducible report: accuracy, Brier score and log loss against chance and rules of thumb

LIMITS AND TAKEAWAYS

What this can and cannot say

- One person, synthetic cities, five proxy metrics: nothing here generalises to real places or other people.
- Weights describe associations in the pairs I chose to compare, not values or expertise.
- Small samples give wide intervals; a model that ties a one-line rule has not shown more than that rule.
- The value is the workflow: making trade-offs explicit, recording real choices, and testing a model honestly before believing it.

The most useful number on these pages is the one where the simple rule did just as well.

